

STABLE RECOVERY GUARANTEES FOR BLIND DECONVOLUTION UNDER RANDOM MASK ASSUMPTION

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ABSTRACT. This study addresses the blind deconvolution problem with modulated inputs, focusing on a measurement model where an unknown blurring kernel $\mathbf{h}$ is convolved with multiple random modulations $\{\mathbf{d}_l\}_{l=1}^L$ (coded masks) of a signal $\mathbf{x}$, subject to ℓ_2 -bounded noise. We introduce a more generalized framework for coded masks, enhancing the versatility of our approach. Our work begins within a constrained least squares framework, where we establish a robust recovery bound for both $\mathbf{h}$ and $\mathbf{x}$, demonstrating its near-optimality up to a logarithmic factor. Additionally, we present a new recovery scheme that leverages sparsity constraints on $\mathbf{x}$. This approach significantly reduces the sampling complexity to the order of $L = O(\log n)$ when the non-zero elements of $\mathbf{x}$ are sufficiently separated. Furthermore, we demonstrate that incorporating sparsity constraints yields a refined error bound compared to the traditional constrained least squares model. The proposed method results in more robust and precise signal recovery, as evidenced by both theoretical analysis and numerical simulations. These findings contribute to advancing the field of blind deconvolution and offer potential improvements in various applications requiring signal reconstruction from modulated inputs.

1. INTRODUCTION

1.1. Problem Setup. Blind deconvolution is an inverse problem that aims to reconstruct two unknown signals, $\mathbf{h}, \mathbf{x} \in \mathbb{C}^n$, from their circular convolution $\mathbf{y} \in \mathbb{C}^n$, defined as $\mathbf{y} := \mathbf{h} \circledast \mathbf{x}$, where $\circledast$ denotes the circular convolution operator. This operation can be equivalently expressed in matrix form as $\mathbf{y} = \mathbf{h} \circledast \mathbf{x} = \mathbf{C}_{\mathbf{h}} \mathbf{x}$, where $\mathbf{C}_{\mathbf{h}}$ is the circulant matrix generated by $\mathbf{h} = [h_1, \dots, h_n]^T$, defined as:

$$\mathbf{C}_{\mathbf{h}} = \begin{bmatrix} h_1 & h_n & \cdots & h_2 \\ h_2 & h_1 & \cdots & h_3 \\ \vdots & \vdots & \ddots & \vdots \\ h_n & h_{n-1} & \cdots & h_1 \end{bmatrix}.$$

This problem arises in numerous fields, including astronomy, optics, image processing, and communications engineering [17, 11, 21, 33]. The blind deconvolution problem is inherently ill-posed due to the presence of scaled-shift symmetry, which implies that there are infinitely many signal pairs that can yield the same convolution result. Consequently, incorporating prior information is crucial to overcoming this ill-posedness. For example, one might impose a subspace condition on $\mathbf{x}$ [2], or enforce a short support condition on $\mathbf{h}$ in combination with a sparsity constraint on $\mathbf{x}$ [19, 31].

In this work, we examine a related class of blind deconvolution problems, where the blur kernel $\mathbf{h} \in \mathbb{C}^n$ is convolved with multiple modulated inputs. Specifically, the observations $\mathbf{y}_l \in \mathbb{C}^n$,

2020 *Mathematics Subject Classification.* Primary 94A15, 46C05; Secondary 94A12, 49N45.

Key words and phrases. Self calibration, General random mask, Optimal complexity, Alternating minimization.

Song Li is supported by NSFC grant (U21A20426, 12071426).

Yu Xia was supported by NSFC grant (12271133, 11901143) and the key project of Zhejiang Provincial Natural Science Foundation grant (LZ23A010002).